

RAS Prelims 2026 · Handwritten Class Notes

Geography of the World

*Chapter-by-chapter notes in coaching-lecture style —
concept first, then the facts, with flow charts, mind
maps, previous-year callouts and current-affairs links.*

WG01 *Physical Geography: Mountains, Plateaus, Plains, Deserts*

WG02 *Rivers and Lakes of the World*

WG03 *Natural Vegetation and Biomes*

WG04 *World Agriculture*

WG05 *World Industry, Transport and Environmental Issues*

*RPSC RAS & RTS Preliminary Examination · 6 December 2026 · 150 questions · 200 marks · 1/3
negative marking*

WG01

Physical Geography: Mountains, Plateaus, Plains, Deserts

About 1-2 questions a paper. Two things repeat: highest peak by continent (RAS 2016, 2018, 2021) and desert matched to continent (RAS 2024). Everything else is a single one-liner.

World Geography carries only about 3 questions in the whole paper, and this chapter supplies most of them. Almost nothing here is analytical - it is peaks, deserts, winds and currents matched to places. Learn it as pairs, not as prose.

Physical Geography of the World

Interior & tectonics

- Moho
- Gutenberg
- Pangaea
- Ring of Fire

Mountains

- Everest
- Aconcagua
- Denali
- Kilimanjaro
- Elbrus
- Vinson

Plateaus & plains

- Tibet
- Colorado
- Katanga
- Anatolian
- Great Plains

Deserts

- Sahara
- Gobi
- Atacama
- Kalahari
- Great Victoria
- Patagonian

Landforms

- Barchan
- Yardang
- Cirque
- Fjord
- Drumlin

Air & ocean

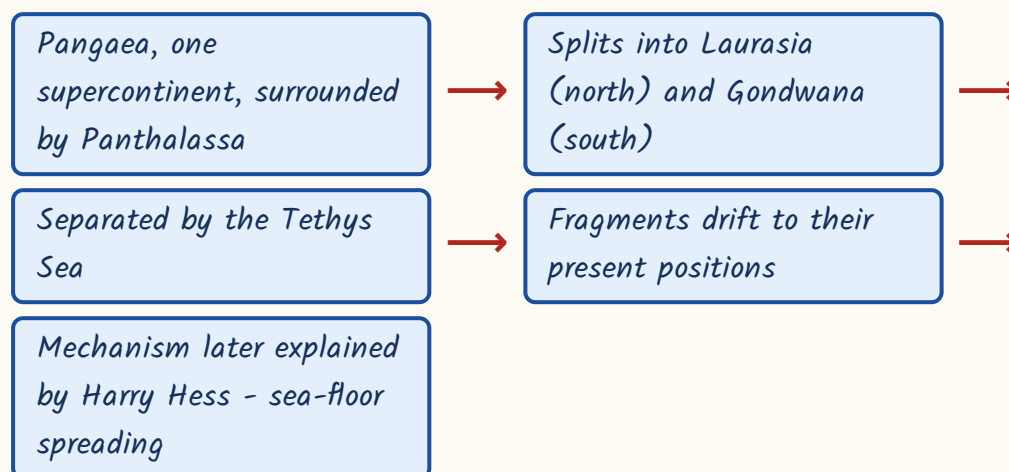
- Pressure belts
- Local winds
- Currents
- Tides
- Coral reefs

Inside the earth, and why the continents move

Start with what the earth is made of, because the discontinuities and the wave behaviour are asked as straight one-liners. The earth has three shells - crust, mantle, core. The surfaces where one shell ends and the next begins are named after the scientists who found them. We know these boundaries only because earthquake waves change speed and direction there.

- Crust-mantle boundary = Mohorovicic discontinuity (Moho).
- Mantle-core boundary = Gutenberg discontinuity.
- By composition the layers are called SIAL (crust), SIMA (mantle), NIFE (core).
- P-waves (primary) pass through solids and liquids both.
- S-waves (secondary) travel only through solids.
- The S-wave shadow zone beyond 103 degrees proves the outer core is liquid.

Continental drift - Wegener, 1912



- Circum-Pacific belt = the 'Ring of Fire' - about two-thirds of the world's active volcanoes.
- It is dominated by subduction zones, so it also has the great majority of earthquakes.
- Ojos del Salado (Chile-Argentina) - world's highest volcano.
- Mauna Loa (Hawaii) - largest active volcano.

- Etna – Europe's most active volcano; Stromboli – 'Lighthouse of the Mediterranean'.

YAAD RAKHO

Moho is the mantle's roof, Gutenberg is the core's gate. Roof first, gate deeper – that fixes the order for you.

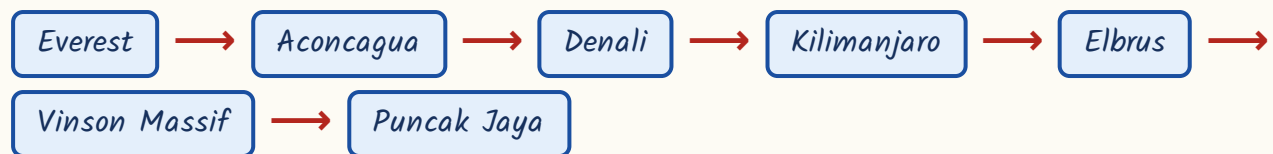
Mountains – the seven summits and the ranges

This is the single most repeated part of world geography in RAS. The question comes in three shapes: which is the highest peak of continent X, match peaks to continents, and arrange peaks in descending order of height. One table answers all three. Learn the peak, the continent and the range together.

Continent to highest peak

Continent	Highest peak	Height	Range / country
Asia (and the world)	Mount Everest	8,848.86 m	Great Himalaya, Nepal-China border
South America	Aconcagua	about 6,960 m	Andes, Argentina
North America	Denali (formerly Mt McKinley)	6,190 m	Alaska Range, USA
Africa	Mount Kilimanjaro	5,895 m	Tanzania; dormant stratovolcano
Europe	Mount Elbrus	5,642 m	Caucasus, Russia
Antarctica	Vinson Massif	4,892 m	Sentinel Range
Oceania	Puncak Jaya (Carstensz Pyramid)	4,884 m	New Guinea, Indonesia

Descending order of continental highest peaks



- K2 / Godwin Austen (8,611 m), Karakoram - world's second highest peak.
- Kangchenjunga (8,586 m) - third highest.
- Mount Kenya (5,199 m) - Africa's second highest.
- Andes (about 7,000 km) - longest mountain range on land.
- Mid-Oceanic Ridge (about 65,000 km) - longest range on earth overall, but under the sea.
- Ural Mountains - conventional boundary between Europe and Asia.
- Atlas - north-west Africa; Drakensberg - South Africa; Great Dividing Range - eastern Australia.

ASKED IN RAS

The direct form. RAS Pre 2018 - which is the highest peak of the African continent? Kilimanjaro. RAS Pre 2016 - Aconcagua lies in which range? The Andes. RAS Pre 2021 - Vinson Massif is the highest peak of which continent? Antarctica. Same table, asked three different years.

ASKED IN RAS

The twisted form. RAS Pre 2021 also asked the peaks arranged in descending order of height, and RAS Pre 2016 asked which peak-continent pair is NOT correctly matched - the wrong pair was 'Mount Kosciuszko - Europe'. So learn the table in both directions and learn the order, not just the pairs.

TRAP

Three confusions RPSC exploits here. (1) Elbrus is Europe's highest; Mont Blanc (4,808 m, Alps) is only the highest in Western Europe. (2) Puncak Jaya is the highest of Oceania; Kosciuszko (2,228 m) is only the highest on mainland Australia - and it is in Australia, never Europe. (3) Everest is the highest peak, but the Andes, not the Himalaya, is the longest range on land.

Plateaus and plains

A plateau is an upland with a flat top standing above the land around it. The Tibetan Plateau is the one that gets named - highest and largest in the world, average elevation above 4,500 m, lying between the Himalaya and the Kunlun, which is why it is called the 'Roof of the World'. For the exam what matters is the plateau's type and its country.

Plateau to type and country

Plateau	Type / feature	Country
Tibetan	Intermontane; 'Roof of the World'	China
Bolivian	Intermontane	Bolivia
Anatolian	Intermontane	Turkey
Deccan	Lava / volcanic	India
Columbia	Lava / volcanic	USA
Colorado	Dissected - the Grand Canyon is cut into it by the Colorado river	USA
Katanga (Shaba)	Copper-rich	DR Congo
Meseta	Continental block	Spain
Mato Grosso	Continental block	Brazil

Plateau	Type / feature	Country
Loess Plateau	Wind-deposited silt	China
Patagonian	Rain-shadow plateau east of the Andes	Argentina
Chotanagpur	Continental plateau	India

- Plateau types to recall: intermontane, lava, dissected, continental, piedmont, rain-shadow.
- Plains are classified as structural, depositional (alluvial, flood, delta, loess) and erosional (peneplain).
- Named plains: Great Plains (North America), North European Plain, West Siberian Plain.
- Also the Amazon plain (South America), the Pampas plain (Argentina) and the Indo-Gangetic plain (India).
- Depositional plains carry the world's densest populations - they are flat, watered and fertile.

Deserts of the world - the matching RPSC actually asked

RAS Pre 2024 put its world-geography matching question on deserts. That makes this the highest-value table in the chapter. Do not learn deserts as a list; learn each desert with its continent, and then with its country. A desert is any area with very low precipitation - which is why the polar ice sheets count as deserts too.

Deserts of Africa and Asia

Desert	Continent	Country / region
Sahara	Africa	North Africa - the largest hot desert, over 8.5 m sq km
Kalahari	Africa	Southern Africa - Botswana, Namibia, South Africa

Desert	Continent	Country / region
Namib	Africa	Coastal Namibia - the oldest desert
Gobi	Asia	Mongolia and northern China
Arabian (with the Rub al-Khali)	Asia	Arabian peninsula
Karakum	Asia	Turkmenistan
Kyzylkum	Asia	Uzbekistan - Kazakhstan
Taklamakan	Asia	Xinjiang, China
Thar	Asia	India - Pakistan
Kavir	Asia	Iran

Deserts of the Americas and Australia

Desert	Continent	Country / feature
Atacama	South America	Chile - the driest desert on earth
Patagonian	South America	Argentina - largest desert of the Americas
Great Basin, Sonoran, Mojave, Chihuahuan	North America	USA and Mexico
Great Victoria	Australia	Largest desert in Australia
Great Sandy	Australia	Western Australia

- The two largest deserts on earth are polar: Antarctic (about 13.9 m sq km) and Arctic (about 13.7 m sq km).
- Sahara is the largest HOT desert - that qualifier decides the answer.
- Atacama is kept dry by the cold Peru current offshore and the rain shadow of the Andes.
- Namib is the oldest desert; Patagonian is a rain-shadow desert east of the Andes.

ASKED IN RAS

RAS Pre 2024 asked deserts three ways in the same rotation - the Gobi is spread over Mongolia and northern China; the Patagonian desert lies mainly in Argentina; and a NOT-correctly-matched pair in which 'Namib - South America' was the wrong one. Namib is in southern Africa.

YAAD RAKHO

Four pairs carry the marks - G-P-V-K. Gobi = Asia, Patagonia = South America, Victoria = Australia, Kalahari = southern Africa. And one rule that explains the west-coast deserts: a cold current offshore means a desert on the coast - Peru current with Atacama, Benguela with Namib, Canary with the western Sahara.

Landforms of erosion and deposition

Wind shapes the deserts, ice shapes the high mountains and high latitudes. RPSC asks these as 'which one of these is NOT a glacial landform' - so you need to know which agent made which feature, not the process. Remember one thing above all: yardang is wind, not ice.

Wind (desert) landforms vs glacial landforms

Wind / desert	Glacier / ice
Erosion: yardang (wind-fluted ridge), zeugen, inselberg, ventifact, mushroom rock, pediment	Erosion: cirque, arete, horn, U-shaped valley, hanging valley, fjord (a drowned glacial valley invaded by the sea)
Deposition: barchan (crescent dune, horns pointing downwind), seif or longitudinal dune, loess (wind-blown silt)	Deposition: moraine, drumlin, esker, outwash plain

CURRENT LINK

Current link: the United Nations declared 2025 the International Year of Glaciers' Preservation, and 21 March is now observed every year as World Day for Glaciers (UNESCO and WMO). Glacier retreat is the standard current-affairs peg for cirques, moraines and fjords - as of September 2026 this remains the live campaign.

Atmosphere, winds, currents and the sea

The atmosphere is layered by how temperature behaves with height. All weather sits in the lowest layer; the ozone that protects us sits in the next one up. Above that, pressure belts and the winds between them set the world's climatic pattern - and the named local winds are a favourite matching set.

- Upward order: troposphere - stratosphere - mesosphere - thermosphere (ionosphere) - exosphere.
 - Troposphere holds all weather, water vapour and dust; 18 km thick at the equator, only 8 km at the poles.
 - Normal lapse rate in the troposphere - 6.5 degrees C fall per km of ascent.
 - Ozone layer lies in the stratosphere, not the troposphere.
 - Mesosphere is the coldest layer; meteors burn up here.
-
- Four pressure belts: equatorial low (doldrums), sub-tropical high (horse latitudes), sub-polar low, polar high.
 - Planetary winds blow from high to low pressure: Trade winds, Westerlies, Polar Easterlies.
 - The Westerlies of the Southern Ocean are the 'Roaring Forties'.
 - Local winds are named, seasonal and regional - they are what get matched in the exam.

Local winds - name, home and nature

Local wind	Where it blows	Nature
Chinook	Eastern slopes of the Rockies	Warm and dry - the 'snow-eater'
Foehn	Alps	Warm and dry
Mistral	Rhone valley, France	Cold
Bora	Adriatic coast	Cold
Sirocco	Sahara to the Mediterranean	Hot and dust-laden
Harmattan	West Africa	Dry; called 'the Doctor' for the relief it brings
Santa Ana	California	Hot and dry
Zonda	Argentina	Warm and dry, off the Andes
Loo	North India	Hot and dry summer wind

Ocean currents - warm against cold

Warm currents	Cold currents
Gulf Stream	Labrador
North Atlantic Drift	Canary
Kuroshio	Oyashio
Brazil	Peru (Humboldt)
Agulhas	Benguela
East Australian	California
-	West Wind Drift

- Spring tides occur on new moon and full moon, when sun, earth and moon are in line (syzygy).
- Neap tides occur at quadrature, when the sun and moon pull at right angles - the weakest tides.
- Bay of Fundy, Canada - the world's highest tidal range.
- Coral reef types: fringing, barrier and atoll.

- *Great Barrier Reef (Coral Sea, off Australia) - the world's largest coral reef system.*

REVISION RECAP

1. Moho separates crust from mantle; Gutenberg separates mantle from core; S-waves stop at the liquid outer core.
2. Wegener 1912 - Pangaea in Panthalassa, splitting into Laurasia and Gondwana across the Tethys; Hess explained the mechanism.
3. Ring of Fire = Circum-Pacific belt = about two-thirds of active volcanoes; Ojos del Salado is the highest volcano.
4. Everest > Aconcagua > Denali > Kilimanjaro > Elbrus > Vinson Massif > Puncak Jaya.
5. Elbrus is Europe's highest; Mont Blanc only Western Europe's; Kosciuszko only mainland Australia's.
6. K2 (8,611 m) second in the world, Kangchenjunga (8,586 m) third; Andes longest range on land, Mid-Oceanic Ridge longest overall.
7. Tibet - highest and largest plateau; Colorado - dissected, Grand Canyon; Katanga - copper, DR Congo; Meseta - Spain.
8. Largest deserts overall are the Antarctic and Arctic polar deserts; Sahara is only the largest hot desert.
9. Gobi-Asia, Patagonia-South America, Great Victoria-Australia, Kalahari-southern Africa: the 2024 matching set.
10. Atacama driest (cold Peru current plus Andean rain shadow); Namib oldest; both lie on west coasts.
11. Yardang, zeugen, barchan and loess belong to wind; cirque, arete, horn, fjord, drumlin and esker belong to ice.
12. Troposphere has all the weather, ozone sits in the stratosphere, mesosphere is the coldest layer.
13. Chinook and Foehn are warm; Mistral and Bora are cold; Harmattan is 'the Doctor'; Sirocco is dusty.
14. Warm - Gulf Stream, Kuroshio, Brazil, Agulhas; cold - Labrador, Canary, Peru, Benguela, Oyashio.
15. Spring tide at syzygy, neap tide at quadrature; Bay of Fundy highest range; Great Barrier Reef largest reef.

WG02

Rivers and Lakes of the World

About 1 question a paper out of the 3-question world geography slot. It is nearly always a match-the-following or a NOT-correctly-matched pair. Canals, straits and lake superlatives are the safest bets.

This chapter is pure matching - river to continent, river to outfall, city to river, lake to superlative, strait to the two land masses it separates. There is nothing to understand and everything to fix in memory. Read the tables aloud twice; that is how these marks are won.

Rivers & Lakes of the World

Longest

- Nile
- Amazon
- Yangtze
- Mississippi-Missouri
- Yenisei

Outfalls

- Volga-Caspian
- Danube-Black Sea
- Congo-Atlantic
- Huang He-Bohai

Cities on rivers

- Paris-Seine
- London-Thames
- Rome-Tiber
- Berlin-Spree

Waterfalls

- Angel
- Victoria
- Niagara
- Iguazu
- Boyoma

Canals & straits

- Suez
- Panama
- Kiel
- Bering
- Malacca
- Hormuz

Lakes

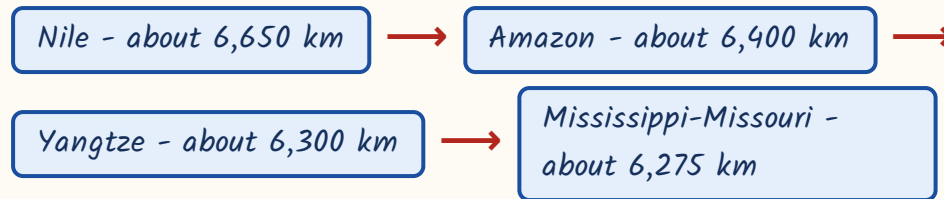
- Caspian
- Superior
- Baikal
- Victoria
- Titicaca
- Dead Sea

Length, discharge, basin - three different questions

Before the tables, settle one confusion. 'Longest' is about length. 'Largest' can mean discharge or basin area. The Nile is the longest river of the world; the Amazon is the largest by both volume of water and basin area. If a question

says 'carries the largest volume of water to the sea', the answer is the Amazon, never the Nile.

Descending order of river length



- Nile - longest in the world and in Africa; rises in the East African highlands via the Kagera.
- Amazon - rises in the Peruvian Andes, drains to the Atlantic; largest drainage basin on earth.
- Yangtze - longest river of Asia and the longest flowing wholly within one country.
- Congo (about 4,700 km) - the only major river that crosses the Equator twice; the world's deepest river.
- Volga (about 3,645 km) - longest river of Europe, entirely inside Russia.
- Danube - second longest in Europe; passes through 10 countries and 4 capital cities.

YAAD RAKHO

For the length order remember N-A-Y-M: Nile, Amazon, Yangtze, Mississippi-Missouri. Four numbers all between 6,200 and 6,700 km - so learn the order, not the figures.

River to continent to outfall - the master tables

This is the table the matching question is built from. Do it continent by continent. Notice the pattern in Siberia - every big Russian river of Asia runs

north into the Arctic. Notice too that two giants, the Congo and the Amazon, end in estuaries and not deltas.

Africa and the Americas

River	Continent	Drains into
Nile	Africa	Mediterranean Sea, through a large delta in Egypt
Congo	Africa	Atlantic Ocean, through an estuary
Niger	Africa	Gulf of Guinea
Zambezi	Africa	Indian Ocean; forms the Victoria Falls
Orange	Africa	Atlantic Ocean
Limpopo	Africa	Indian Ocean
Amazon	South America	Atlantic Ocean, through an estuary
Parana - Rio de la Plata (about 4,880 km)	South America	Atlantic Ocean; second longest of the continent
Orinoco	South America	Atlantic Ocean, Venezuela
Sao Francisco	South America	Atlantic Ocean, Brazil
Mississippi-Missouri	North America	Gulf of Mexico, bird's-foot delta
Mackenzie	North America	Beaufort Sea (Arctic)
Yukon	North America	Bering Sea
Colorado	North America	Gulf of California

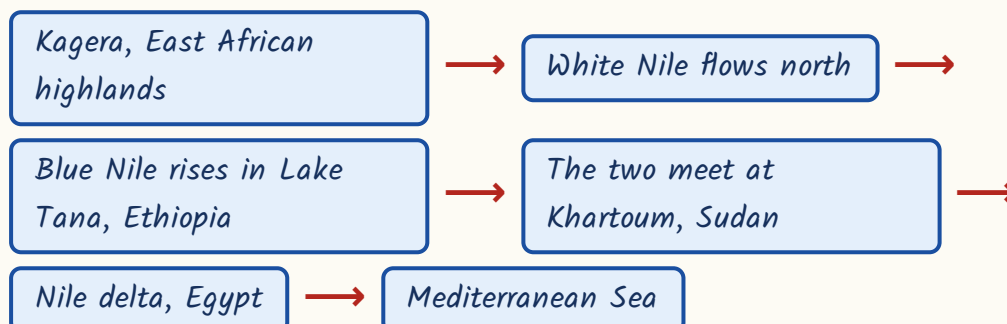
Asia, Europe and Australia

River	Continent	Drains into
Yangtze	Asia	East China Sea

River	Continent	Drains into
Huang He (Yellow)	Asia	Bohai Sea
Mekong	Asia	South China Sea
Ob-Irtysh (5,410 km)	Asia	Gulf of Ob, Arctic
Yenisei (5,539 km)	Asia	Kara Sea, Arctic
Lena (4,400 km)	Asia	Laptev Sea, Arctic
Tigris and Euphrates	Asia	Persian Gulf, after uniting as the Shatt al-Arab in southern Iraq
Volga	Europe	Caspian Sea
Danube	Europe	Black Sea
Rhine	Europe	North Sea, at Rotterdam
Rhone	Europe	Mediterranean Sea
Murray-Darling	Australia	Southern Ocean; longest river system of Australia

- Huang He - the 'Sorrow of China' because of its floods; carries a huge load of loess.
- St. Lawrence drains the Great Lakes into the Atlantic.
- Mekong flows through China, Myanmar, Laos, Thailand, Cambodia and Vietnam.
- Ganga-Brahmaputra delta (the Sundarbans) is the world's largest delta.

Course of the Nile - the confluence question



TRAP

Delta or estuary? The Congo and the Amazon discharge through estuaries, not true deltas, even though they are enormous. The Mississippi has a bird's-foot delta and the Ganga-Brahmaputra has the largest delta of all.

Cities on rivers

Every capital in Europe grew on a river. RPSC picks four of them and asks you to match, or gives one wrong pair and asks which is wrong. The Danube alone carries four capitals - that fact by itself answers several questions.

City to river

City	River	Country
Paris	Seine	France
London	Thames	United Kingdom
Rome	Tiber	Italy
Berlin	Spree	Germany
Hamburg	Elbe	Germany
Warsaw	Vistula	Poland
Rotterdam	Rhine	Netherlands
Vienna	Danube	Austria
Bratislava	Danube	Slovakia
Budapest	Danube	Hungary
Belgrade	Danube	Serbia
Cairo	Nile	Egypt
Khartoum	Blue Nile and White Nile meet here	Sudan
Baghdad	Tigris	Iraq

TRAP

Paris stands on the Seine, not the Rhine. That exact swap is the standard wrong option. Similarly Hamburg is on the Elbe, Rotterdam on the Rhine - do not merge the two German cities.

Waterfalls, deltas and confluences

Waterfalls are asked as height, river or country. Only four names really matter: Angel is the highest, Boyoma has the greatest volume, Victoria is the famous one on the Zambezi, and Niagara sits on the USA-Canada border.

Waterfall to river and country

Waterfall	River	Country
Angel (979 m) - world's highest	Falls from the Auyan-tepui into a tributary of the Churun	Venezuela (Canaima National Park)
Tugela - second highest	Tugela	South Africa
Victoria - 'Mosi-oa-Tunya', the smoke that thunders	Zambezi	Zambia - Zimbabwe border
Niagara	Niagara	USA - Canada
Iguazu	Iguazu	Argentina - Brazil
Boyoma - greatest volume of any waterfall	Congo	DR Congo

- Angel Falls is a plunge of 979 m - the height figure is the examinable bit.
- Victoria Falls lies on the Zambezi, which then flows on to the Indian Ocean.
- Iguazu and Niagara are both border falls - remember the country pairs.
- Boyoma wins on discharge, not height.

ASKED IN RAS

RPSC's world geography slot is a matching slot - RAS Pre 2016, 2018 and 2021 used mountain peaks, RAS Pre 2024 used deserts. Rivers and waterfalls sit in the same rotation, so treat 'Angel-Venezuela', 'Victoria-Zambezi' and 'Boyoma-Congo' as pairs to be recalled instantly rather than as facts to be reasoned out.

Canals and straits

Two canals are asked again and again, and the distinction between them is the whole question: Suez has no locks, Panama has locks. Learn the years, the lengths and the water bodies each one joins.

Suez Canal vs Panama Canal

Suez Canal	Panama Canal
Opened 1869	Opened 1914
About 193 km long	About 82 km long
Joins the Mediterranean (Port Said) to the Red Sea (Suez)	Joins the Atlantic Ocean to the Pacific Ocean
A sea-level canal with no locks	Works through a system of locks that lift ships to Gatun Lake
Nationalised by Egypt in 1956	Cuts the Panama isthmus of Central America

- Kiel Canal - opened 1895, about 98 km, joins the North Sea to the Baltic Sea in Germany.
- Suez shortens the Europe-Asia route by removing the voyage round the Cape of Good Hope.
- Panama removes the voyage round Cape Horn at the tip of South America.

Straits - what they separate and what they connect

<i>Strait</i>	<i>Separates</i>	<i>Connects</i>
<i>Bering</i>	<i>Asia and North America</i>	<i>Pacific and Arctic oceans</i>
<i>Gibraltar</i>	<i>Europe and Africa</i>	<i>Mediterranean Sea and Atlantic Ocean</i>
<i>Malacca</i>	<i>Sumatra and the Malay peninsula</i>	<i>Andaman Sea and South China Sea</i>
<i>Hormuz</i>	<i>Iran and the Arabian peninsula</i>	<i>Persian Gulf and Gulf of Oman</i>
<i>Bab-el-Mandeb</i>	<i>Arabia and Africa</i>	<i>Red Sea and Gulf of Aden</i>
<i>Bosphorus</i>	<i>Asian and European Turkey</i>	<i>Black Sea and Sea of Marmara</i>
<i>Magellan</i>	<i>Mainland South America and Tierra del Fuego</i>	<i>Atlantic and Pacific oceans</i>
<i>Palk Strait</i>	<i>India and Sri Lanka</i>	<i>Palk Bay and the Bay of Bengal side</i>

CURRENT LINK

Current link: the Panama Canal depends on fresh water from Gatun Lake, and the El Nino drought of 2023-24 forced the canal authority to cut daily transits and tighten draft limits; conditions eased through 2025. As of September 2026 this is the standing example of climate stress on a trade route - useful for any canal or shipping question.

ASKED IN RAS

The bank's NOT-correctly-matched question on canals turns on one line: 'Panama Canal - a sea-level canal without locks' is FALSE. It is Suez that has no locks. Expect the same trick with 1869 and 1914 swapped.

Lakes - superlatives and types

Lakes are asked as superlatives. There are four big ones and they are four different lakes - largest, deepest, largest freshwater, highest navigable.

Confusing them is the commonest mistake. After the superlatives, learn how lakes are formed, because 'lake to type' is the matching version of the same question.

Lake superlatives

Superlative	Lake	Detail
Largest lake on earth by area	Caspian Sea	About 3.7 lakh sq km; landlocked salt lake shared by Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan
Largest freshwater lake by area	Lake Superior	82,100 sq km, North America
Deepest lake, and largest freshwater lake by volume	Lake Baikal	1,642 m; tectonic rift lake in Siberia; about a fifth of the world's unfrozen surface fresh water
Second deepest, and longest freshwater lake	Lake Tanganyika	1,470 m, East Africa
Largest lake of Africa	Lake Victoria	About 69,000 sq km; second largest freshwater lake by area
Largest lake of Europe	Lake Ladoga	Russia
Highest navigable lake	Lake Titicaca	3,812 m, on the Peru-Bolivia border
Lowest exposed land surface on earth	Dead Sea shore	About 430 m below sea level
Largest salt lake and lowest point of Australia	Lake Eyre	South Australia

Lakes by type of origin

Type	Examples
Tectonic (rift or downwarped)	Baikal, Tanganyika, Dead Sea
Crater or caldera	Crater Lake (USA), Toba (Indonesia)
Glacial	The Great Lakes of North America
Ox-bow (fluvial)	Cut-off meanders on flood plains

Type	Examples
Lagoon (coastal)	Enclosed by sand bars along the coast
Artificial (reservoir)	Lake Volta (Ghana), Lake Nasser (Egypt)

- Aral Sea (Kazakhstan-Uzbekistan) shrank drastically after the Amu Darya and Syr Darya were diverted for cotton irrigation.
- Lake Chad is shared by Chad, Niger, Nigeria and Cameroon.
- Sargasso Sea in the North Atlantic is the only sea with no coastline - it is bounded by the currents of the North Atlantic gyre.

YAAD RAKHO

Four superlatives, four different lakes: Caspian largest, Baikal deepest, Superior largest freshwater by area, Titicaca highest navigable. And Lake Malawi is NOT the deepest - that wrong option appears in the bank.

ASKED IN RAS

Because the world geography slot rotates through matching sets - peaks in 2016, 2018 and 2021, deserts in 2024 - lakes and straits are the obvious untouched pool. The two sets worth over-learning are the lake superlatives above and the strait table: Bering (Asia-North America), Gibraltar (Europe-Africa), Malacca, Hormuz, Bab-el-Mandeb, Palk.

REVISION RECAP

1. Nile longest (about 6,650 km); Amazon largest in discharge and basin area; Yangtze longest in Asia and within one country.
2. Length order: Nile > Amazon > Yangtze > Mississippi-Missouri.
3. Congo crosses the Equator twice, is the deepest river, and ends in an estuary like the Amazon.
4. Volga - longest in Europe, into the Caspian; Danube - 10 countries, 4 capitals, into the Black Sea.

5. *Siberian rivers run north: Ob-Irtysh to the Gulf of Ob, Yenisei to the Kara Sea, Lena to the Laptev Sea.*
6. *Huang He = 'Sorrow of China', loess-laden, into the Bohai Sea; Mekong into the South China Sea.*
7. *Tigris and Euphrates unite as the Shatt al-Arab and enter the Persian Gulf.*
8. *Blue Nile (Lake Tana) meets the White Nile at Khartoum; Sundarbans is the world's largest delta.*
9. *Cities: Paris-Seine, London-Thames, Rome-Tiber, Berlin-Spree, Hamburg-Elbe, Warsaw-Vistula, Cairo-Nile.*
10. *Angel (979 m, Venezuela) highest; Boyoma greatest volume; Victoria on the Zambezi; Iguazu Argentina-Brazil.*
11. *Suez 1869, 193 km, Mediterranean to Red Sea, no locks; Panama 1914, 82 km, Atlantic to Pacific, locks and Gatun Lake.*
12. *Kiel Canal 1895, 98 km, North Sea to Baltic.*
13. *Straits: Bering Asia-North America, Gibraltar Europe-Africa, Hormuz Persian Gulf-Gulf of Oman, Palk India-Sri Lanka.*
14. *Caspian largest lake, Superior largest freshwater by area, Baikal deepest and largest by volume, Titicaca highest navigable.*
15. *Aral Sea shrank from cotton irrigation on the Amu Darya and Syr Darya; Sargasso is the sea without a coastline.*

WG03

Natural Vegetation and Biomes

About 1 question a paper. Two things are documented repeats: the temperate grassland names matched to their regions, and the RAS 2024 fact that tropical forests record the maximum deforestation in the world.

Biomes is the friendliest chapter in world geography. It is a set of names - Selva, Llanos, Prairie, Steppe, Veld, Taiga, Tundra - each tied to one region and one kind of plant. Fix the names to the map and the questions answer themselves.

World Biomes

Tropical

- Equatorial Selva
- Monsoon deciduous
- Savanna

Temperate grasslands

- Prairie
- Pampas
- Steppe
- Veld
- Downs
- Canterbury
- Puszta

Mediterranean scrub

- Maquis
- Chaparral
- Matorral
- Fynbos
- Mallee

Cold lands

- Taiga - largest land biome
- Tundra - permafrost

Special habitats

- Desert xerophytes
- Mangrove - Sundarbans

Forest facts

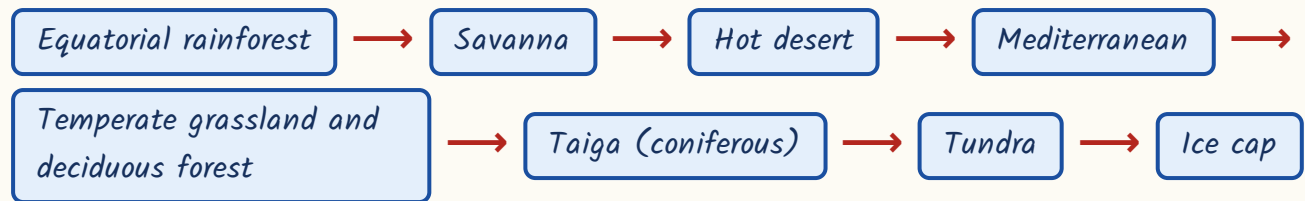
- 4.14 bn ha
- 31-32% of land
- Tropical forests lose most

How the biome map is arranged

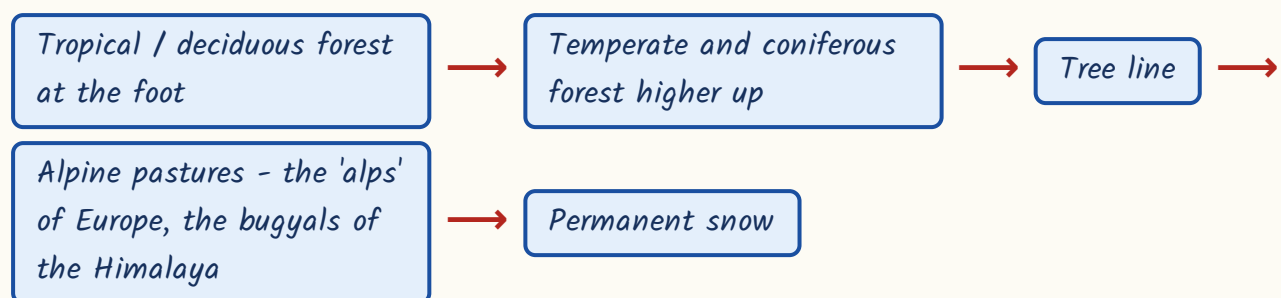
Vegetation follows climate, and climate follows latitude. So the belts run in a fixed order from the Equator to the pole. Learn that order once and you can place any biome. The same sequence repeats vertically as you climb a mountain

- that is why the Himalaya shows tropical forest at the foot and alpine pasture near the top.

Vegetation belts from the Equator poleward



The same sequence going up a mountain (altitudinal zonation)



ASKED IN RAS

RPSC gives world geography about 3 questions and takes them as matching sets - peaks in 2016, 2018 and 2021, deserts and deforestation in 2024. In this chapter the matching pools are exactly three: temperate grassland names, Mediterranean scrub names, and the tropical trio Selva-Llanos-Campos. Over-learn those three lists.

The tropical belt - rainforest, monsoon forest, savanna

Three biomes share the tropics and they differ by how the rain is spread through the year. Rain all year gives evergreen rainforest. Rain in one season gives deciduous monsoon forest. Still less rain gives grassland with scattered trees - the savanna.

- Equatorial evergreen rainforest is locally called 'Selva' in the Amazon.

- It occurs in the Amazon basin, the Congo basin and South-East Asia.
 - Multi-storeyed with a dense canopy, lianas and epiphytes; there is no distinct leaf-shedding season.
 - Hardwoods: mahogany, ebony, rosewood, greenheart, cinchona.
 - Trees grow scattered in mixed stands, so commercial exploitation is difficult.
-
- Tropical deciduous or monsoon forest - teak, sal, sandalwood, bamboo.
 - It sheds its leaves in the dry season to save water.
 - Commercially it is the most useful forest of the monsoon lands, including India.
-
- Savanna is tropical grassland - tall coarse grass with scattered acacia and baobab.
 - It is called the 'big game country' - lion, zebra, giraffe, elephant, wildebeest.
 - Do not confuse the names: Selva is a forest, while Llanos and Campos are grasslands.

Tropical grassland names by region

Name	Region
Llanos	Orinoco basin, Venezuela
Campos	Brazilian highlands
Sudan / Sahel type savanna	Africa
Northern Australian savanna	Australia

Temperate grasslands - the repeat you cannot miss

This is the highest-yield table in the chapter. Every temperate grassland has a local name and one region, and the question is always the pairing. These grasslands lie in continental interiors, get only 25-75 cm of rain, and are

practically treeless - which is why they became the world's granaries and its great ranching lands.

Temperate grassland to region

Grassland	Region
Prairie	North America - USA and Canada
Pampas	Argentina and Uruguay
Steppe	Eurasia - Russia, Ukraine, Central Asia
Veld	South Africa
Downs	Australia
Canterbury	New Zealand
Puszta	Hungary
Manchurian	North-east China

- Practically treeless, because rainfall is too low and too seasonal for trees.
- They lie in continental interiors, far from the moderating sea.
- Rainfall 25-75 cm a year.
- They are the world's granaries - wheat and maize - and its great ranching lands.
- The herdsmen of the Pampas are called gauchos.

YAAD RAKHO

Walk the world in order: Prairie (North America) - Pampas (South America) - Steppe (Eurasia) - Veld (Africa) - Downs (Australia) - Canterbury (New Zealand) - Puszta (Hungary). Same first letters as the continents in most cases: P-P-S-V-D-C-P.

ASKED IN RAS

The question bank marks *Prairie-North America*, *Pampas-Argentina*, *Veld-South Africa* and *Canterbury-New Zealand* as repeat items, and the NOT-correctly-matched version uses 'Pampas - South Africa' as the wrong pair. RAS asks this as a straight match-the-following, so recall must be instant.

TRAP

The two swaps RPSC lives on: *Veld* is *South Africa* and *Pampas* is *Argentina* - never the other way round. And *Downs* is the *Australian grassland*, not the chalk downs of England.

Mediterranean lands and temperate deciduous forest

Mediterranean climate means hot dry summers and mild wet winters - the reverse of the monsoon. Plants survive the dry summer by being sclerophyllous: thick bark, small waxy leaves, deep roots. These are the orchard lands of the world, and each region gives its scrub a different name.

Mediterranean scrub names

Scrub name	Region
<i>Maquis and Garrigue</i>	<i>Mediterranean basin</i>
<i>Chaparral</i>	<i>California</i>
<i>Matorral</i>	<i>Central Chile</i>
<i>Fynbos</i>	<i>Cape region, South Africa</i>
<i>Mallee</i>	<i>South-western Australia</i>

- Typical crops: citrus fruit, olives, grapes and cork oak.
- The region is called the 'orchard land' of the world.

- Adaptations to the dry summer: thick bark, small waxy leaves, deep roots.
- Temperate deciduous forest - oak, ash, beech, elm, maple.
- Found in western Europe, eastern USA and eastern Asia.
- It is the most cleared of all forest belts, because these are the world's old industrial and farming regions.

Cold lands and special habitats

North of the deciduous belt come two biomes that are always asked against each other - taiga and tundra. Taiga is the coniferous forest, tundra is the treeless cold desert beyond the tree line. Both lie almost entirely in the Northern Hemisphere, simply because there is very little land at those latitudes in the south.

Taiga vs Tundra

Taiga (boreal coniferous forest)	Tundra
Largest land biome; a continuous belt across Siberia, Canada and Scandinavia	Lies beyond the tree line, on the Arctic fringe
Conifers - pine, fir, spruce, larch, cedar; evergreen, conical, needle-leaved	Mosses, lichens (reindeer moss) and sedges; no trees
World's chief source of softwood, pulp, newsprint and matchwood	No commercial timber; growing season only about two months
Grizzly bear, moose, lynx, wolf	Caribou or reindeer, musk ox, arctic fox, polar bear
Confined almost wholly to the Northern Hemisphere	Underlain by permafrost - permanently frozen sub-soil

- Desert vegetation is xerophytic - cactus, acacia, date palm, euphorbia.
- Xerophyte adaptations: long roots, thick cuticle, spines in place of leaves, water-storing stems.

- Mangrove or tidal forest is salt tolerant, with stilt roots and pneumatophores for breathing.
- The Sundarbans of the Ganga-Brahmaputra delta is the world's largest mangrove forest.
- Orangutan - Borneo and Sumatra; gorilla, okapi and bonobo - Congo basin.
- Grizzly bear, moose, lynx and wolf - North American coniferous belt.
- Kangaroo - Australian grassland.
- Coniferous softwood forests are the world's leading lumbering regions.
- Equatorial hardwood forests are lumbered far less - mixed stands and poor accessibility.

YAAD RAKHO

Taiga has the timber, tundra has the turf. Both begin with T; the one with trees is the one whose name has more letters in common with 'timber'.

Forest statistics and deforestation - the 2024 question

This is the one part of the chapter with hard numbers, and RAS 2024 asked it twice in the same paper - once directly and once as a NOT-correct statement. The single fact to burn in: tropical forests record the maximum deforestation in the world. Not temperate, not boreal - tropical.

- World forest area is about 4.14 billion hectares - roughly 31-32 per cent of the global land area.
- More than half the world's forests lie in five countries: Russia, Brazil, Canada, the United States and China.
- Russia alone has about 832 million hectares - about one-fifth of the world total.
- Nearly half of the world's forests are tropical.

- The Amazon, largely in Brazil, holds the world's largest area of tropical rainforest and is called the 'lungs of the earth'.
- Global deforestation has slowed - from about 17.6 million hectares a year in the 1990s to about 10.9 million hectares a year in the most recent decade.
- The largest direct driver of tropical deforestation is conversion of forest to agricultural land.
- That means shifting cultivation, cattle ranching in the Amazon, and oil-palm and soy plantations in South-East Asia and Brazil.

ASKED IN RAS

RAS Pre 2024 asked outright: at present the maximum deforestation in the world is taking place in which type of forest? Answer - tropical forests. The same paper's negative version made 'Temperate forests are recording the maximum deforestation in the world' the incorrect statement. One fact, two questions.

CURRENT LINK

Current link: the FAO's Global Forest Resources Assessment 2025, released in October 2025, confirms world forest area at about 4.14 billion hectares and annual forest loss down to about 10.9 million hectares a year for 2015-25, from 17.6 million in the 1990s; South America still loses the most forest each year. As of September 2026 this is the latest FAO assessment - quote its figures, not older ones.

TRAP

Two number traps. Forests cover about 31 per cent of the LAND area, which is roughly one-third of the land - not one-third of the earth's surface, because most of the earth is ocean. And deforestation is slowing, but it has not stopped; do not read the falling rate as forest gain.

REVISION RECAP

1. Selva = equatorial rainforest of the Amazon; Llanos = Orinoco grassland; Campos = Brazilian highlands grassland.
2. Equatorial hardwoods - mahogany, ebony, rosewood, greenheart, cinchona; scattered stands make logging hard.
3. Monsoon deciduous forest - teak, sal, sandalwood, bamboo; commercially the most useful of the monsoon lands.
4. Savanna = tall coarse grass with acacia and baobab; the 'big game country'.
5. Prairie-North America, Pampas-Argentina, Steppe-Eurasia, Veld-South Africa, Downs-Australia, Canterbury-New Zealand, Puszta-Hungary.
6. Temperate grasslands are treeless, continental, 25-75 cm rain, the world's granaries; Pampas herdsmen are gauchos.
7. Mediterranean scrub: Maquis and Garrigue in the basin, Chaparral in California, Matorral in Chile, Fynbos in the Cape, Mallee in south-west Australia.
8. Mediterranean lands are the world's orchards - citrus, olives, grapes, cork oak; sclerophyllous plants.
9. Taiga is the largest land biome, almost wholly in the Northern Hemisphere, and supplies the world's softwood.
10. Tundra lies beyond the tree line on permafrost - mosses, lichens, sedges; growing season about two months.
11. Temperate deciduous forest (oak, ash, beech, elm, maple) is the most cleared forest belt of all.
12. Sundarbans is the world's largest mangrove; mangroves use stilt roots and pneumatophores.
13. World forests about 4.14 billion ha, 31-32 per cent of land; Russia, Brazil, Canada, USA and China hold over half.
14. TROPICAL forests record the maximum deforestation - the RAS 2024 fact, asked directly and negatively.
15. Deforestation fell from about 17.6 m ha a year in the 1990s to about 10.9 m ha a year now; the driver is farmland conversion.

WG04

World Agriculture

About 1 question a paper. The one documented repeat is Mediterranean agriculture and its five regions (RAS 2024, asked twice in that paper, once in the 'which is NOT' form). Everything else is a single one-liner or a match-the-following.

World Geography gives you only about three questions in the whole paper, and agriculture is one of the two places RPSC reliably digs into. Nothing here is analytical. It is types of farming matched to regions, and crops matched to leading producers. Learn it as pairs.

World Agriculture

Subsistence

- Shifting (Jhum, Milpa, Ladang)
- Nomadic herding (Bedouin, Masai)
- Intensive subsistence (monsoon Asia)

Collective / co-op

- Kolkhoz
- Sovkhoz
- Denmark

Commercial

- Extensive grain (Prairie, Pampas, Steppe)
- Mixed farming
- Dairy
- Plantation
- Market gardening

Crops & producers

- Rice/wheat - China
- Maize - USA
- Coffee & sugarcane - Brazil
- Cocoa - Cote d'Ivoire
- Palm oil - Indonesia

Mediterranean

- Med. basin
- California
- Central Chile
- SW Cape
- SW Australia

Fisheries

- NW Pacific
- Grand Banks
- Dogger Bank
- Peru upwelling

What 'type of agriculture' means - the classification RPSC tests

Agriculture is classified by one simple question: who eats the produce? If the farmer and his family eat most of it, that is subsistence farming. If it goes to the market, that is commercial farming. Everything else - farm size, labour,

machinery, yield - follows from that one split. RPSC never asks you to argue this. It asks you to place a farming type in a region, or a region in a farming type.

- Subsistence: shifting cultivation, nomadic herding, intensive subsistence (wet paddy).
- Commercial: extensive grain, mixed farming, dairy, plantation, Mediterranean, market gardening.
- State-controlled: collective farming (Kolkhoz) and co-operative farming (Denmark).
- Yield per hectare and output per worker are two different things - keep them separate.
- Intensive subsistence has high yield per hectare but very low output per worker.
- Extensive commercial grain farming is the reverse - low yield per hectare, huge output per worker.

Von Thunen's isolated-state model, 1826 - rings outward from the market



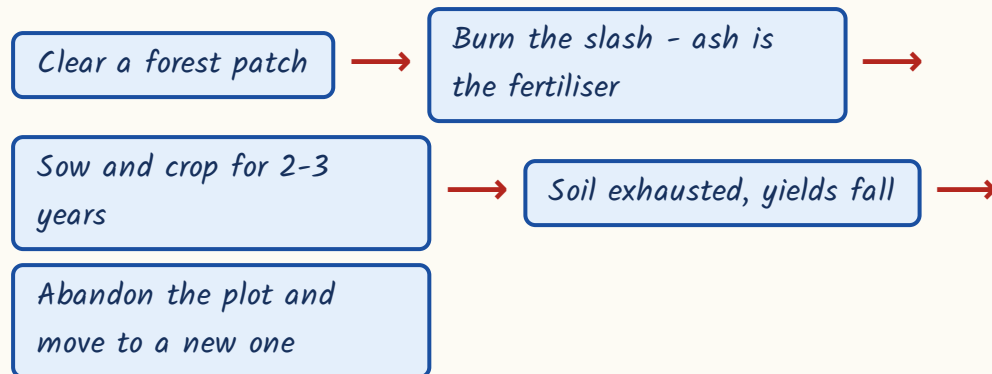
YAAD RAKHO

Von Thunen's logic in one line: the more perishable and the heavier the produce, the closer it must sit to the city. Milk and vegetables first, cattle last.

Subsistence farming - shifting, nomadic, intensive

Shifting cultivation is the oldest of the three. You clear a patch of forest, burn it, sow on the ash, take two or three crops until the soil is exhausted, then abandon the plot and move on. It survives in tropical forests. The examinable part is not the method - it is the long list of local names, which comes as a match-the-following.

The shifting cultivation cycle



Shifting cultivation - local names

Local name	Where
Jhum	North-east India
Milpa	Mexico and Central America
Ladang	Indonesia and Malaysia
Chena	Sri Lanka
Roca	Brazil
Ray	Vietnam
Taungya	Myanmar
Conuco	Venezuela
Masole	Congo
Humah	Java

Nomadic herders - community to region and animal

Community	Region	Chief animal
Bedouin	Arabia and the Sahara	Camel

Community	Region	Chief animal
Masai	East Africa	Cattle
Lapps (Sami)	European tundra	Reindeer
Chukchi	Siberian tundra	Reindeer
Kirghiz and Kazakh	Central Asian steppe	Horse, sheep
Fulani	West Africa	Cattle

- Intensive subsistence farming = monsoon Asia, wet paddy dominant.
- Holdings are tiny, the family supplies the labour, machinery is minimal.
- Yield per hectare is high because of heavy manuring and multiple cropping.
- Output per worker stays very low - too many hands on too little land.

TRAP

Do not confuse nomadic herding with transhumance. Nomadic herders have no permanent home and wander the arid lands. Transhumance is the seasonal up-and-down movement between highland summer pasture and lowland winter pasture - the Gujjars and Bakarwals of the western Himalayas do this, and they do have fixed settlements.

Commercial farming - grain, dairy, plantation, market gardening

Commercial farming means growing for sale. In the temperate grasslands it takes the extensive form - enormous farms, heavy machinery, wheat and cattle. In the tropics it takes the plantation form - one cash crop, a large estate, hired labour. Near cities it becomes market gardening. The grassland names are the classic matching set, so learn the grassland with its country.

Extensive commercial grain farming - grassland to country

Grassland	Country / region
Prairie	USA and Canada
Pampas	Argentina
Steppe	Russia and Ukraine
Veld	South Africa
Downs	Australia
Canterbury	New Zealand

- Plantation agriculture was introduced in the tropics largely by European settlers.
- One cash crop, large estate, hired labour, heavy capital in processing and transport.
- Rubber - Malaysia; tea - Sri Lanka; coffee - Brazil; cocoa - Ghana; banana - Central America.
- It does NOT grow a variety of food crops for local consumption - that is the standard 'NOT a feature' option.

Plantation vs intensive subsistence

Plantation	Intensive subsistence
Tropics, colonial origin	Monsoon Asia, indigenous
Very large estates	Very small holdings
Hired wage labour	Family labour
Single cash crop for export	Food crops for the household
High capital, processing on site	Low capital, hand tools

- Commercial dairy farming is best developed in north-western Europe, north-eastern USA and New Zealand.
- Denmark is the classic case of co-operative farming - the 'Dairy of Europe'.
- New Zealand is the largest exporter of dairy products; India is the largest producer of milk.

- Market gardening or truck farming = intensive growing of vegetables, fruit and flowers close to big cities.
- 'Factory farming' = stall-fed livestock reared indoors on bought feed.
- Kolkhoz = the collective farm of the erstwhile USSR, run jointly by member households; collectivisation began in 1929.
- Sovkhoz = the state farm, owned by the state and worked by wage labour.
- Collective farming is compulsory pooling; co-operative farming (Denmark) is voluntary.

Mediterranean agriculture – the RPSC repeat, learn it cold

This is the highest-value half page in the chapter. Mediterranean agriculture occurs only where the Mediterranean climate occurs – hot dry summers, mild wet winters – and that is exactly five belts in the world, all on the western side of continents between roughly 30 and 40 degrees latitude. RPSC asked it in 2024 and it asked it in the negative form: which region is NOT associated with Mediterranean agriculture. So you must know the five, and you must know the common wrong options.

The five Mediterranean agriculture belts

Belt	Continent / country
Mediterranean basin	Southern Europe, North Africa, the Levant
Central California	USA
Central Chile	South America
South-western Cape	South Africa
South-western and southern Australia	Australia

- Called the 'orchard lands of the world' – citrus, olives, figs, grapes.
- Viticulture, that is grape growing for wine, is their most distinctive commercial activity.

- Winter rain suits cereals; the long dry summer suits deep-rooted fruit trees and vines.

ASKED IN RAS

Asked in RAS Pre 2024 - which region is NOT associated with the Mediterranean type of agriculture? The answer was the Pampas of Argentina. The Pampas is a temperate grassland of extensive grain farming and cattle ranching, not a Mediterranean belt.

ASKED IN RAS

Also asked in RAS Pre 2024 - Mediterranean agriculture is characteristically associated with the cultivation of citrus fruits, olives and grapes. Two questions from one topic in one paper. That is how narrow this section is.

TRAP

The trap options are always other famous farming regions - Pampas, Prairie, Steppe, Downs. All four are temperate grasslands, none of them is Mediterranean. Remember: Mediterranean belts sit on WEST coasts in the 30-40 degree band.

Crops and their leading producers

Learn producer and exporter separately, because RPSC's favourite trap is to swap them. A country can grow the most of something and still export almost none of it, because its own people eat it. India and milk is the obvious case. Stick to the stable leaders below - do not chase ranks that move a place or two every season.

Leading producers - food crops

Crop	No. 1 producer	No. 2
Rice	China	India
Wheat	China	India
Maize	USA	China
Soybean	Brazil	USA
Sugarcane	Brazil (about 40 per cent of world output)	India
Potato	China	India
Groundnut	China	India
Milk	India	-

Leading producers - beverage, industrial and tree crops

Crop	No. 1 producer	Point to remember
Coffee	Brazil	Vietnam No. 2 and the largest Robusta producer; coffee originated in the Ethiopian highlands
Tea	China	India No. 2, Kenya No. 3, Sri Lanka No. 4
Cocoa	Cote d'Ivoire	Ghana No. 2; both on the Gulf of Guinea coast
Natural rubber	Thailand	Indonesia No. 2, Vietnam No. 3; Hevea brasiliensis is native to the Amazon
Palm oil	Indonesia	With Malaysia No. 2, about 85 per cent of world output
Olives / olive oil	Spain	Italy No. 2
Jute	India	But Bangladesh is the largest exporter
Fish	China	Capture plus aquaculture, and largest exporter too

- Largest exporters: rice - India; wheat - Russia; maize - USA and Brazil; soybean - Brazil.
- Wool - Australia is the largest exporter.
- Dairy products - New Zealand is the largest exporter.
- Kenya is the largest exporter of black tea, though China and India produce far more.

- ~~Cotton is the one big crop whose leading-producer rank keeps moving among China, India, Brazil and the USA - learn the group, not a No. 1.~~

TRAP

Producer is not exporter. India produces the most milk and exports little. India produces the most jute but Bangladesh exports the most. China produces the most wheat but Russia exports the most. RPSC has built whole 'NOT correctly matched' questions on exactly this gap.

ASKED IN RAS

Worth knowing what World Geography actually looked like in RAS Pre 2024 - only three questions, and two of them were pure matching: deserts to continents, Mediterranean agriculture regions, and tropical forests recording the highest deforestation. Read that as instruction: memorise pairs, do not read theory.

Fishing grounds, the revolutions, and FAO

The world's great fisheries sit where cold and warm waters meet, or where cold nutrient-rich water rises from below. That mixing brings up plankton and the fish follow. Two names come up again and again - the Grand Banks and the Peruvian anchoveta - so fix the location and the cause for each.

Major fishing grounds

Ground	Location	Why it is rich
North-West Pacific	Off Japan, Korea, eastern Russia	World's most productive fishing ground
North-East Atlantic	Off north-west Europe	Broad continental shelf

Ground	Location	Why it is rich
Grand Banks	Off Newfoundland, Canada	Labrador Current meets the Gulf Stream
Dogger Bank	North Sea	Shallow shelf bank
Georges Bank	Off New England, USA	Shelf and current mixing
Peru-Chile coast	West coast of South America	Upwelling of the Peru (Humboldt) Current

- The Peruvian anchoveta fishery depends on upwelling of cold, nutrient-rich water.

- In El Nino years the upwelling weakens and the catch collapses.

- China is the largest producer of fish - capture plus aquaculture - and the largest exporter of fishery products.

- Norman Borlaug - father of the Green Revolution, Nobel Peace Prize 1970, for dwarf high-yielding wheat.

- M. S. Swaminathan led the Indian adaptation of it from 1966-67.

- Colour revolutions: Green - foodgrains, White - milk, Blue - fish, Yellow - oilseeds, Golden - horticulture and honey.

- Pink - prawn and onion, Silver - eggs, Red - meat and tomato, Brown - leather and cocoa, Round - potato.

- White Revolution = Operation Flood, led by Verghese Kurien; it made India the largest milk producer.

- FAO was founded on 16 October 1945, headquarters Rome; World Food Day is 16 October.

CURRENT LINK

Current link, as of September 2026: the FAO has designated 2026 the International Year of the Woman Farmer, launched to push gender equality in agrifood systems. RPSC likes UN 'International Year' designations in the current-affairs block - pair it with FAO's founding date, 16 October 1945, and World Food Day.

REVISION RECAP

1. Subsistence vs commercial - the only split that matters; everything else follows from it.
2. Von Thunen 1826: dairy and market gardening innermost, ranching outermost.
3. Shifting cultivation names: Jhum-NE India, Milpa-Mexico, Ladang-Indonesia, Chena-Sri Lanka, Roca-Brazil, Taungya-Myanmar.
4. Herders: Bedouin-camel, Masai-cattle, Lapps and Chukchi-reindeer, Fulani-West Africa.
5. Transhumance is seasonal highland-lowland movement, not nomadism; Gujjar-Bakarwal is the Indian example.
6. Grain grasslands: Prairie-USA/Canada, Pampas-Argentina, Steppe-Russia/Ukraine, Veld-South Africa, Downs-Australia, Canterbury-New Zealand.
7. Plantation = one cash crop, big estate, hired labour, European origin in the tropics.
8. Kolkhoz collective, Sovkhoz state farm, Denmark co-operative.
9. Mediterranean agriculture = five belts only: Med. basin, central California, central Chile, SW Cape, SW/southern Australia. Citrus, olives, grapes; viticulture is the signature.
10. Rice, wheat, tea, potato, groundnut, fish - China No. 1. Maize - USA. Soybean, sugarcane, coffee - Brazil.
11. Cocoa - Cote d'Ivoire; rubber - Thailand; palm oil - Indonesia; olives - Spain; milk and jute - India.
12. Exporters: rice-India, wheat-Russia, dairy-New Zealand, wool-Australia, jute-Bangladesh, black tea-Kenya.
13. Grand Banks off Newfoundland - Labrador Current meets Gulf Stream. NW Pacific is the most productive ground.
14. Peru anchoveta lives on upwelling; El Nino kills the catch.
15. Borlaug - Green Revolution, Nobel Peace 1970. FAO 16 October 1945, Rome. 2026 is the International Year of the Woman Farmer.

WG05

World Industry, Transport and Environmental Issues

About 1-2 questions a paper. Documented repeats: industrial region to country (RAS 2013, 2016, 2023), greenhouse gases in descending order of contribution (2016, 2018), the acid rain pollutant pair (2016, 2018), biodiversity hotspots (2016, 2018), tropical deforestation (2024).

This is the densest chapter in World Geography and the one RPSC has raided most often. Two things repeat by name: industrial regions matched to countries, and the environment treaties with their years. The environment half also overlaps with the Science and Technology block, so a page learnt here pays twice.

World Industry, Transport & Environment

Europe

- Ruhr - Germany
- Lorraine - France
- Midlands - UK
- Donbas - Ukraine
- Silesia - Poland

Asia & America

- Keihin/Kanto - Japan
- Great Lakes & New England - USA
- Silicon Valley
- Anshan & Shanghai - China

Transport

- Trans-Siberian
- Canadian Pacific
- Trans-Australian
- Suez
- Panama
- Pan-American Highway

Climate regime

- UNFCCC 1992
- Kyoto 1997
- Paris 2015
- COP26-COP31
- IPCC 1988

Ozone & pollution

- Vienna 1985
- Montreal 1987
- Kigali 2016
- Acid rain SO₂ + NO_x
- Plastics treaty

Land & life

- UNCCD 1994
- Ramsar 1971
- CITES 1973
- CBD 1992
- 36 hotspots

Industrial regions matched to countries – the most repeated item in World Geography

Start here and give it the most time. RPSC has asked industrial-region matching in 2013, 2016 and 2023, sometimes as a match-list and sometimes as 'which pair is NOT correctly matched'. The logic behind every one of these regions is the same - coal, or iron ore, or a port. You do not need the logic to score; you need the pair. Learn the region with its country first, then the leading city.

Industrial region to country - learn this table cold

Industrial region	Country	Basis / leading cities
Ruhr	Germany	Coal-based iron and steel; Essen, Dortmund, Duisburg
Saar	Germany	The other German coalfield
Lorraine	France	'Minette' iron ore, Western Europe's largest ore reserve
Midlands / Black Country	United Kingdom	Birmingham, engineering
Donbas (Donetsk Basin)	Ukraine	Coal and steel
Kuzbas (Kuznetsk Basin)	Russia	Coal; also the Urals with Magnitogorsk
Silesia	Poland	Katowice
Sambre-Meuse	Belgium	Coalfield belt
Po valley	Italy	Northern Italy's manufacturing core
Keihin (Kanto plain)	Japan	Tokyo-Yokohama, Japan's largest belt
Great Lakes / Detroit	USA	Automobiles
New England	USA	Earliest centre of cotton textiles
Silicon Valley	USA	California - electronics and IT
Anshan	China	Manchuria, steel

- *United Kingdom in detail: Lancashire (Manchester, cotton - 'Cottonopolis'), Yorkshire (Leeds-Bradford, wool), Clydeside (Glasgow, shipbuilding), Sheffield (cutlery and special steels).*

- Japan in detail: Keihin (Tokyo-Yokohama, largest), Hanshin (Osaka-Kobe), Chukyo (Nagoya), Kitakyushu/Yawata (steel).
- USA in detail: Pittsburgh-Lake Erie is the steel belt; New England is textiles; Silicon Valley is electronics.
- China in detail: Shanghai is the largest industrial city; Wuhan and Beijing-Tianjin are the other cores.

ASKED IN RAS

This one topic has been asked three times. RAS Pre 2013 - which pair of industrial region and country is NOT correctly matched? The wrong pair was Donbas - Poland (Donbas is Ukraine; Silesia is the Polish one). RAS Pre 2016 and again in 2023 - a match-list of Lorraine, Ruhr, Donbas and Midlands against France, Germany, Ukraine and the United Kingdom. And RAS Pre 2016 asked it plainly: Essen, Dortmund and Duisburg belong to the Ruhr.

TRAP

Three pairs RPSC swaps deliberately. Donbas is Ukraine, not Poland. Kuzbas is Russia, not Ukraine. Lorraine is the iron ORE field of France while the Ruhr is the COAL field of Germany - ore in France, coal in Germany.

Minerals, steel and energy - who leads what

This section is a list, and RPSC asks it as a list. Keep to the leaders that have been stable for years. Notice one pattern: the biggest producer of a mineral is often not the biggest exporter of it, because a large economy consumes its own output.

Mineral and energy leaders

Item	Leader	Note
Crude steel	China	More than half of world output; India No. 2, Japan No. 3
Iron ore	Australia	Largest producer and exporter; Brazil No. 2 (Carajas, Itabira)
Copper	Chile	Chuquibambilla and Escondida; also largest reserves
Uranium	Kazakhstan	-
Coal and gold	China	-
Nickel	Indonesia	-
Bauxite reserves	Guinea	Largest reserves in the world
Crude oil	USA	Largest producer; Saudi Arabia is the largest exporter

- OPEC was founded at Baghdad in 1960; its headquarters are at Vienna.
- Steel needs coal plus iron ore plus a cheap route between them - that is why every classic steel region sits on a coalfield or a waterway.
- Duisburg on the Rhine is the world's great inland port and is what makes the Ruhr work.

Transport - railways, the great road, canals, ports

Trans-continental railways were built to bind a large country together, so each one runs coast to coast or capital to far edge. Three names come up. Learn each with its country and its two terminal cities - RPSC's format is 'which pair is NOT correctly matched'.

Trans-continental railways and the longest road

Line	Country	Route and fact
Trans-Siberian Railway	Russia	Moscow to Vladivostok, about 9,289 km - the longest railway line in the world; completed 1916; crosses eight time zones

Line	Country	Route and fact
Trans-Australian (Indian Pacific)	Australia	Perth to Sydney; the world's longest dead-straight stretch of track, about 478 km across the Nullarbor Plain
Canadian Pacific Railway	Canada	Montreal to Vancouver
Pan-American Highway	Alaska to southern Argentina	The longest motorable road; broken only by the Darien Gap

Suez vs Panama - the classic two-canal question

Suez Canal	Panama Canal
Opened 1869	Opened 1914
About 193 km	About 82 km
Port Said to Suez	Atlantic to Pacific
Links the Mediterranean with the Red Sea	Links the Atlantic with the Pacific
Sea-level, NO locks	Has locks - Gatun, Pedro Miguel, Miraflores, plus Gatun Lake
Nationalised by Egypt in 1956	-

- Kiel Canal joins the North Sea to the Baltic.
- The Grand Canal of China is the world's longest canal.
- Shanghai is the world's busiest container port; Singapore second; Rotterdam the largest in Europe.
- Hartsfield-Jackson Atlanta is the busiest airport by total passengers.
- Dubai is the busiest by international passengers; Hong Kong is the busiest for air cargo.

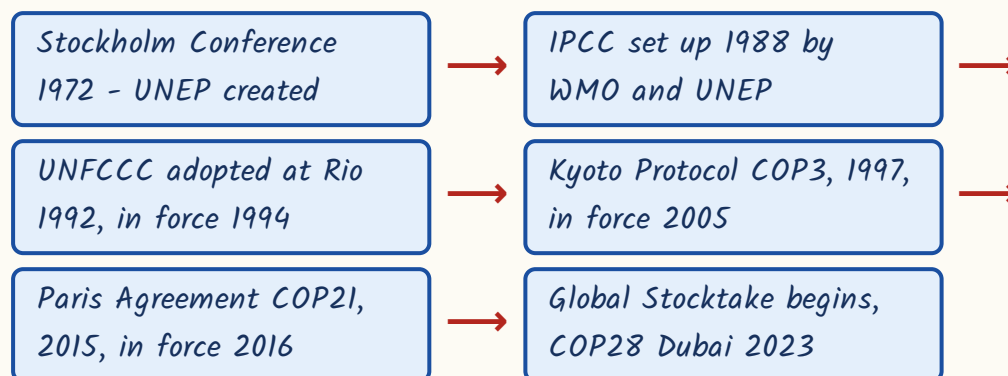
YAAD RAKHO

Suez has No locks because it is at Sea level - S with S. Panama has locks because it must lift ships over the isthmus into Gatun Lake.

Global warming and the climate regime - UNFCCC to Paris to the COPs

Understand the architecture before the dates and the whole thing becomes easy. The UNFCCC of 1992 is the parent convention - it only set the goal and the principle. Under it sit two operative agreements: the Kyoto Protocol of 1997, which put binding cuts on developed countries only, and the Paris Agreement of 2015, under which every country sets its own target. The COP is simply the annual meeting of the parties to the UNFCCC, numbered in sequence.

The climate regime in order



- Stockholm Conference on the Human Environment 1972 created UNEP, headquarters Nairobi, and gave us World Environment Day on 5 June.
- UNFCCC rests on the principle of common but differentiated responsibilities.
- IPCC was set up in 1988 by the World Meteorological Organization and UNEP; it shared the 2007 Nobel Peace Prize with Al Gore; the AR6 Synthesis Report appeared in 2023.
- Kyoto Protocol - adopted at COP3 in 1997, in force 16 February 2005, first commitment period 2008-12.
- Kyoto's three flexible mechanisms: Clean Development Mechanism, Joint Implementation, Emissions Trading.
- Doha Amendment 2012 extended Kyoto to 2020.

- *Paris Agreement* - adopted at COP21 on 12 December 2015, in force 4 November 2016.
- *Paris goal*: hold warming well below 2 degrees Celsius and pursue 1.5 degrees, through Nationally Determined Contributions and a global stocktake every five years.
- *Kyoto basket of gases*: carbon dioxide, methane, nitrous oxide, HFCs, PFCs and sulphur hexafluoride; nitrogen trifluoride was added by the Doha Amendment.
- Sulphur hexafluoride has the highest global warming potential of the basket.
- CFCs are NOT in the Kyoto basket - they are handled by the Montreal Protocol.

COP sequence - hosts and headline outcome



Net-zero target years and India's NDC

Country / bloc	Net-zero year
India	2070

Country / bloc	Net-zero year
China	2060
European Union, United Kingdom, United States, Japan	2050

- India's updated NDC - cut the emission intensity of GDP by 45 per cent by 2030 over 2005 levels.
- India's updated NDC - reach 50 per cent non-fossil installed power capacity by 2030.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2018 - arrange the greenhouse gases in decreasing order of their conventionally estimated contribution to the enhanced greenhouse effect. The order is carbon dioxide (about 60 per cent), then methane (about 20 per cent), then CFCs (about 14 per cent), then nitrous oxide (about 6 per cent). Also remember nitrogen is NOT a greenhouse gas.

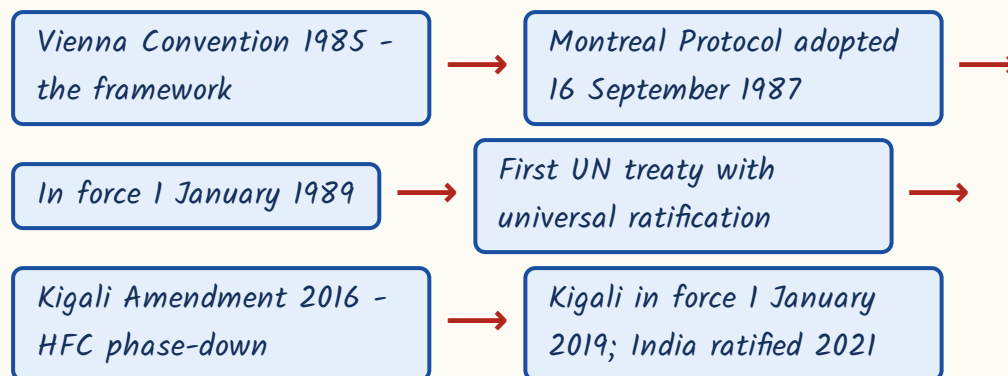
CURRENT LINK

Current link, as of September 2026: COP31 will be held at Antalya, Turkiye, from 9 to 20 November 2026 - after the exam-season news cycle begins. Turkiye hosts and its minister Murat Kurum is COP President, while Australia's Chris Bowen presides over the negotiations under a shared arrangement agreed at COP30. A pre-COP is planned in Fiji in early October 2026. Track it: the host city plus the headline decision is exactly the form RPSC uses.

Ozone, acid rain and pollution

Keep ozone and climate change in separate boxes in your head - RPSC exploits the overlap. Ozone depletion is a stratospheric problem caused by chlorine and bromine compounds, handled by Vienna and Montreal. Global warming is a tropospheric heat problem caused by greenhouse gases, handled by UNFCCC, Kyoto and Paris. The Kigali Amendment is the one bridge between them, because HFCs are greenhouse gases dealt with under an ozone treaty.

The ozone regime



- Ozone lies in the stratosphere and is measured in Dobson Units.
- World Ozone Day, the International Day for the Preservation of the Ozone Layer, is 16 September.
- Kigali was adopted at Kigali, Rwanda, and phases down hydrofluorocarbons - greenhouse gases that do NOT deplete ozone.
- Carbon dioxide is not an ozone-depleting substance; CFCs, halons and carbon tetrachloride are.
- Acid rain - sulphur dioxide and oxides of nitrogen are the precursors; they form sulphuric and nitric acid.
- Rain counts as acid rain when its pH falls below 5.6.
- Damage: the Taj Mahal ('marble cancer'), the Black Forest in Germany, Scandinavian lakes.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2018 - statements on acid rain. Sulphur dioxide and oxides of nitrogen are the principal precursors, and that statement is correct. The statement blaming carbon dioxide is the false one. This pollutant pair is one of the most repeated single facts in the whole environment syllabus.

CURRENT LINK

Current link, as of September 2026: there is still no global plastics treaty. UNEA-5.2 at Nairobi adopted resolution 5/14 'End Plastic Pollution' in March 2022, but the negotiating sessions at Busan in 2024 and Geneva in 2025 both ended without agreement, chiefly on whether to cap plastic production. A short session at Geneva in February 2026 (INC-5.3) only elected a new chair, Julio Cordano of Chile, with substantive talks pushed to a further session later in 2026. Also remember the Great Pacific Garbage Patch lies in the North Pacific gyre between Hawaii and California.

Land, biodiversity and the treaty table

The last section is one table you must be able to write from memory. RPSC asks treaties in two forms - convention matched to year, and convention matched to subject. So learn each treaty as a triplet: year, name, subject. Go down the table in chronological order and it will stick.

Environmental conventions - year, name, subject

Year	Treaty	Subject
1971	Ramsar Convention (Ramsar, Iran)	Wetlands; in force 1975; World Wetlands Day 2 February
1973	CITES (Washington)	Trade in endangered species; in force 1975

Year	Treaty	Subject
1985	Vienna Convention	Framework for protection of the ozone layer
1987	Montreal Protocol	Phase-out of ozone-depleting substances
1989	Basel Convention	Trans-boundary movement of hazardous wastes
1992	CBD (Rio)	Biological diversity
1992	UNFCCC (Rio)	Climate change; in force 1994
1994	UNCCD (Paris, 17 June)	Desertification; in force 1996
1997	Kyoto Protocol	Binding emission cuts for developed countries
1998	Rotterdam Convention	Prior informed consent for hazardous chemicals
2001	Stockholm Convention	Persistent organic pollutants
2013	Minamata Convention	Mercury
2015	Paris Agreement	Climate change; NDCs and the 1.5 degree goal
2016	Kigali Amendment	Phase-down of HFCs

- Under the CBD: Cartagena Protocol on Biosafety 2000, Nagoya Protocol on access and benefit sharing 2010.
- Kunming-Montreal Global Biodiversity Framework - CBD COP15, Montreal 2022, 23 targets including '30x30'.
- World Day to Combat Desertification and Drought is 17 June; UNCCD COP16 met at Riyadh in 2024.
- Land Degradation Neutrality is SDG target 15.3.
- Great Green Wall - launched by the African Union in 2007, an 8,000 km belt of restored land across the Sahel from Senegal to Djibouti.
- Biodiversity hotspots - the concept is Norman Myers's, 1988; there are 36 at present.
- Criteria: at least 1,500 endemic vascular plant species, and loss of at least 70 per cent of the original vegetation.
- India has four - Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland (Nicobar Islands).

- Tropical forests record the highest rate of deforestation of all forest types.

ASKED IN RAS

Asked in RAS Pre 2016 and 2018 - the biodiversity hotspot statements: Norman Myers's concept, the 1,500 endemic plant species threshold, and the 70 per cent vegetation loss criterion were all correct. And in RAS Pre 2024 - tropical forests record the highest deforestation rate among all forest types. Both are worth memorising word for word.

YAAD RAKHO

Two anchors will place most treaties for you. Everything ozone is in the 1980s - Vienna 1985, Montreal 1987. Everything Rio is 1992 - CBD and UNFCCC together. Ramsar 1971 and CITES 1973 sit before them; the chemicals trio Basel-Rotterdam-Stockholm runs 1989-1998-2001.

REVISION RECAP

1. Ruhr-Germany (Essen, Dortmund, Duisburg), Lorraine-France (minette ore), Midlands-UK, Donbas-Ukraine, Kuzbas-Russia, Silesia-Poland.
2. Keihin/Kanto is Japan's largest belt; New England is US textiles, Great Lakes/Detroit is autos, Silicon Valley is electronics.
3. China makes over half the world's crude steel; Australia leads iron ore; Chile copper; Kazakhstan uranium; Guinea bauxite reserves.
4. USA is the largest crude oil producer, Saudi Arabia the largest exporter. OPEC - Baghdad 1960, headquarters Vienna.
5. Trans-Siberian: Moscow-Vladivostok, about 9,289 km, longest in the world, eight time zones.
6. Trans-Australian has the longest straight track (Nullarbor); Canadian Pacific is Montreal-Vancouver; Pan-American Highway breaks at the Darien Gap.
7. Suez 1869, 193 km, no locks. Panama 1914, 82 km, has locks. Kiel joins North Sea to Baltic; Grand Canal of China is the longest canal.
8. Shanghai busiest container port; Atlanta busiest airport by total passengers, Dubai by international, Hong Kong by cargo.

9. UNFCCC 1992 is the parent; Kyoto 1997 binds only developed countries; Paris 2015 runs on NDCs.
10. COP26 Glasgow - India net-zero 2070. COP27 - Loss and Damage Fund. COP28 - first Global Stocktake. COP29 - 300 billion dollars by 2035. COP30 Belem 2025. COP31 Antalya, November 2026.
11. Greenhouse contribution order: CO₂ about 60 per cent > CH₄ about 20 > CFCs about 14 > N₂O about 6. Nitrogen is not a greenhouse gas.
12. Ozone: Vienna 1985, Montreal 1987 (in force 1989, universal ratification), Kigali 2016 for HFCs. World Ozone Day 16 September, Dobson Units, stratosphere.
13. Acid rain = SO₂ + NO_x, pH below 5.6, marble cancer on the Taj.
14. Treaty years: Ramsar 1971, CITES 1973, Basel 1989, CBD and UNFCCC 1992, UNCCD 1994, Rotterdam 1998, Stockholm 2001, Minamata 2013.
15. Hotspots: Myers 1988, 36 today, 1,500 endemic plants plus 70 per cent vegetation loss; India has four. Tropical forests lose the most cover.